Probability of an event

1 State the likelihood that corresponds to each of the following probabilities our of the following: Impossible, Unlikely, Likely or Certain.

a 0.1 **b** 1 **c** 0 **d** 0.6

e $\frac{9}{10}$ **f** $\frac{2}{10}$ **g** 10% **h** 70%

i $\frac{4}{5}$ **j** $\frac{1}{8}$

2 Estimate the probability of the following events:

a It will rain somewhere in Australia sometime this month.

b You meet someone today and correctly guess their birthday.

3 Find the probability of the following events:

a You roll a six-sided die and get 5 or more.

b An event that will never occur.

c An event that will occur 2 times every 50 trials.

d An event that will occur 2 times every 100 trials.

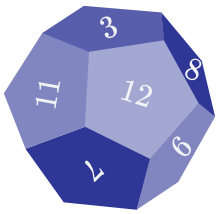
e An event that will occur 2 times every 500 trials.

4 A twelve-sided die is rolled.

a Find the probability of rolling 5 or more.

b Find the probability of rolling less than 5.

c Find the probability of rolling 3 or more.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

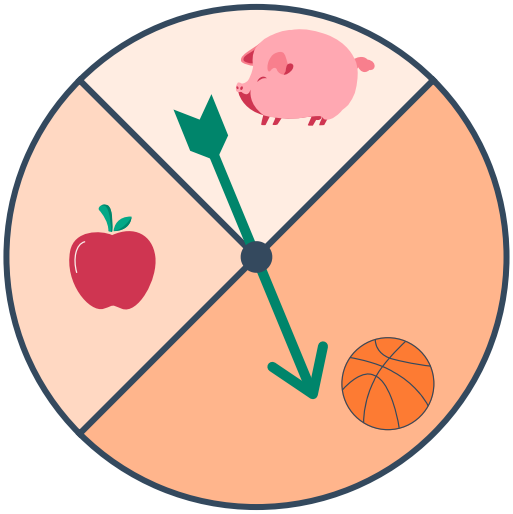
5 Consider the spinner shown:
Find the probability of getting a:

a Apple

b Ball

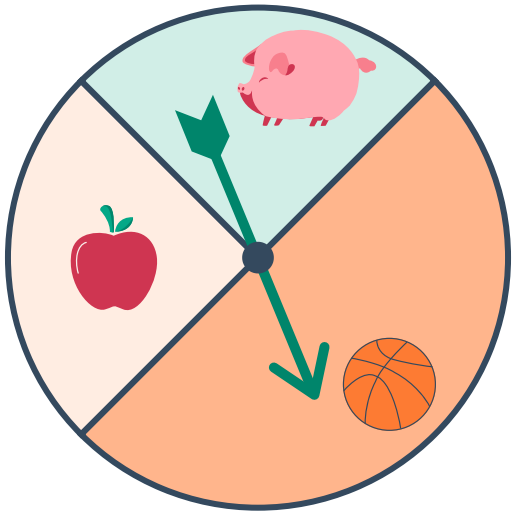
c Pig

d Star

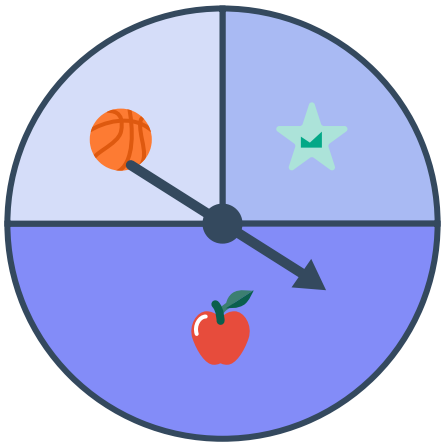


6 For each of the following spinners, find the probability of getting a star. Write your answer as a percentage.

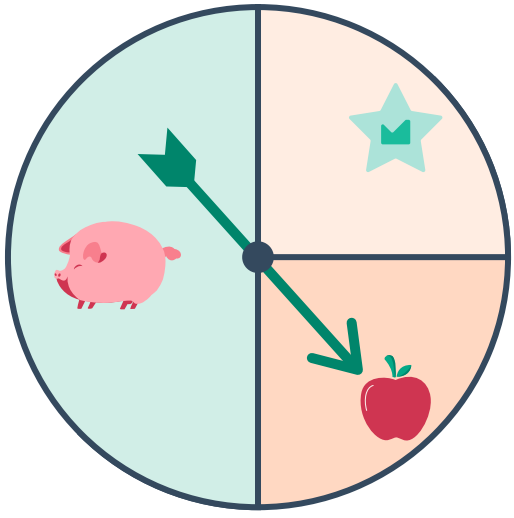
a



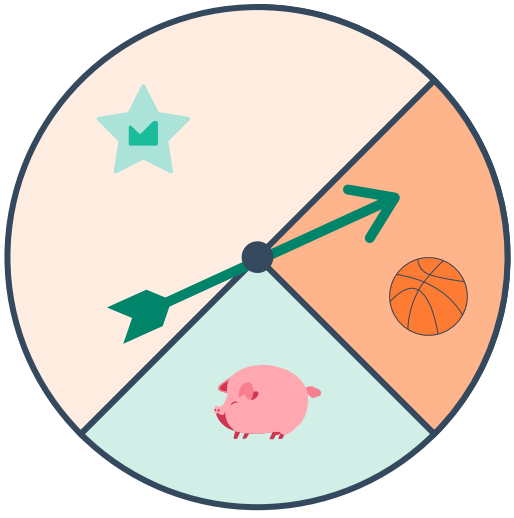
b



c



d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 A jar contains 10 marbles in total. Some of the marbles are blue and the rest are red.

a If the probability of picking a red marble is $\frac{4}{10}$, how many red marbles are there in the jar?

b Find the probability of picking a blue marble.

8 Paul has 23 marbles in a bag and 4 of them are black. Paul picks a marble from the bag without looking. Find the probability that Paul picks a black marble.

9 Jack has a bag of marbles with 5 marbles. 4 of those marbles are red. Find the probability of Jack picking a red marble as a decimal.

10 A book has pages numbered from 1 to 100. If the book is opened to a random page, what is the probability that the page number:

a Is a multiple of 9?

b Has the digit 6 in the page number?

11 Dylan has 32 marbles in a bag. 20 of them are orange. What is the probability Dylan will pick a orange marble from the bag without looking?

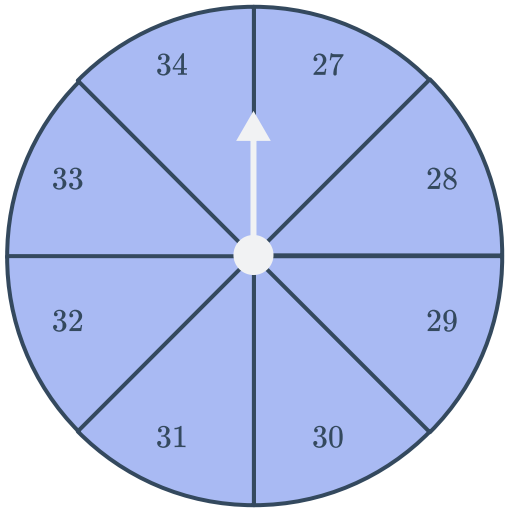
12 A circular spinner is divided into unequal parts. The green sector takes up an angle of 250 degrees at the centre. The red sector takes up an angle of 60 degrees at the centre and the blue sector takes up the remainder of the spinner. What is the probability that the spinner will land on blue?

13 Charlene spins the spinner shown:

a List the sample space.

b Find the probability of spinning a number greater than 29.

c Find the probability of spinning a number in the twenties.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 A number is randomly selected from the following list:

$\{1, 3, 3, 6, 6, 6, 8, 8, 8, 8, 10, 10, 10, 10, 10\}$

a What is the probability of selecting a 1?

b What is the probability of selecting a 3?

c What is the probability of selecting an 8?

d Which number is most likely to be selected?

Events with multiple possibilities

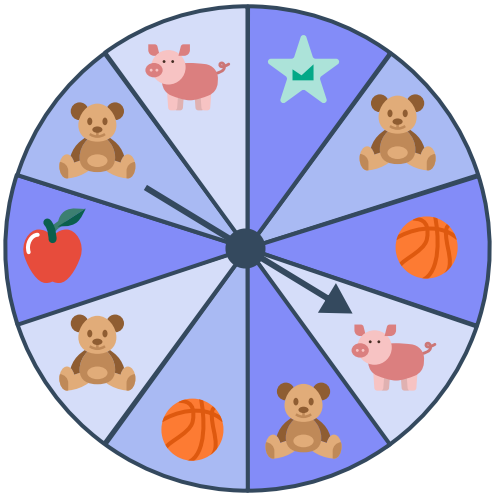
15 Consider the following spinner:
Find the probability of spinning:

a A pig.

b A ball or a pig.

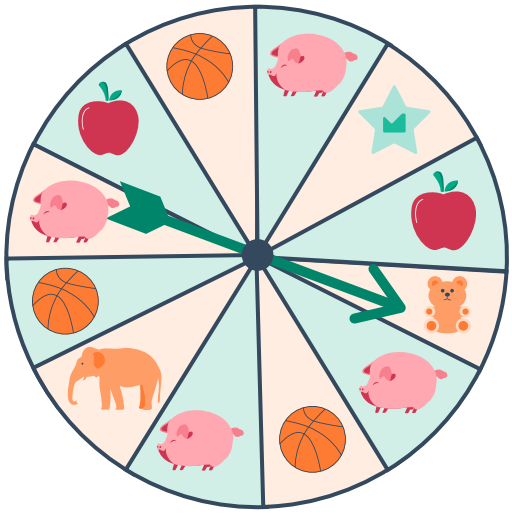
c A pig or an apple.

d An apple or a star.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 Consider the following spinner:



a Find the probability of getting the following shapes. Simplify your answers.

i An apple.

ii A pig.

iii An elephant.

iv A ball.

v A pig or a ball.

vi Anything but an apple.

vii An elephant, a star or a teddy bear.

viii Anything but a ball or teddy bear.

b Determine whether the following events have a probability less than 10%.

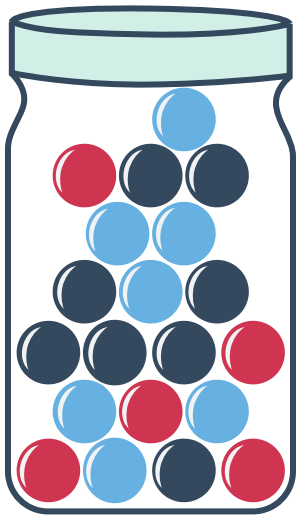
i Spinning a teddy bear.

ii Spinning a star.

iii Spinning an apple.

iv Spinning a ball or an elephant.

17 A single marble is drawn from the following jar 5 red, 7 blue and 8 black marbles:

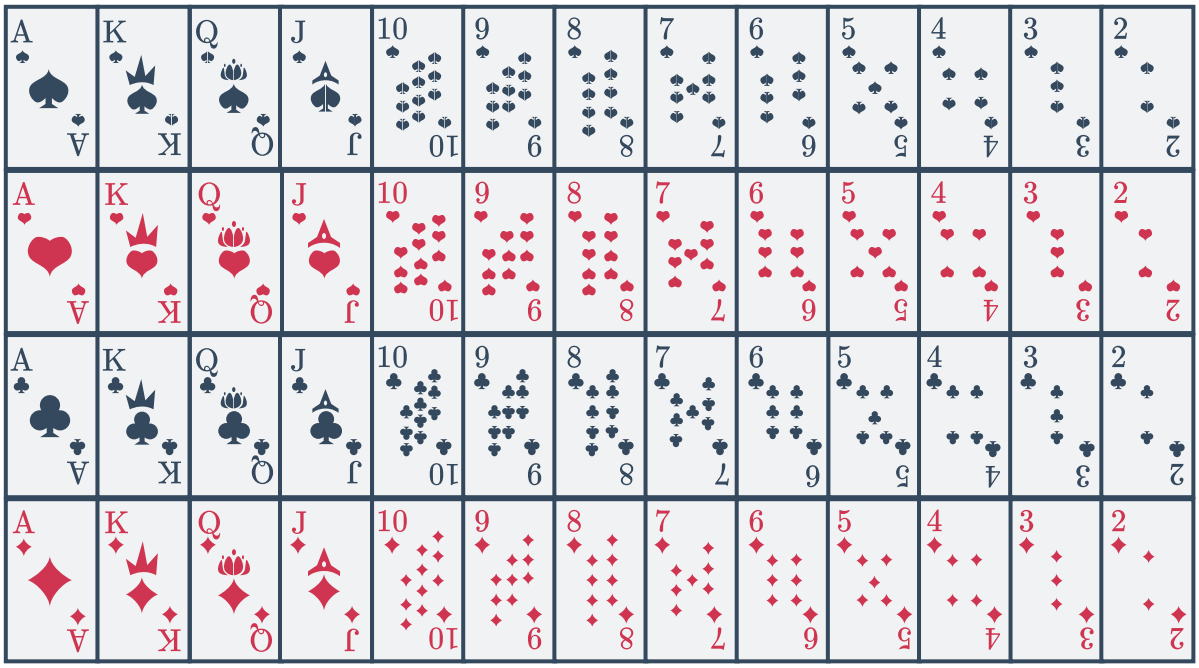


a Find the probability of drawing a blue marble. Give your answer as a percentage.

b Find the probability of drawing a blue marble or a red marble. Give your answer as a percentage.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 A card is drawn from a standard deck of cards:



a Find the probability of drawing the Jack of Hearts or the Jack of Clubs.

b Rosey draws a 6 of Spades and puts it back in the deck and asks you to draw a card. Find the probability that you will draw a card with a number larger than 6, and the same colour as Rosey's card.

19 The 26 letters of the alphabet are written on pieces of paper and placed in a bag. If one letter is to be picked out of the bag at random find the probability of picking a:

a J

b K, Y or R.

c A letter in the word PROBABILITY.

d M, K, D, O, H or B.

e A letter in the word WORKBOOK.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647